

HS AND KESTREL

LET'S UNDERSTAND



Written by
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WHAT IS KESTREL?

Kestrel is a cross-platform web server built into .NET Core. It's lightweight, fast, and runs on Windows, Linux, and macOS.

- Default web server for .NET Core
- **Cross-platform** (Windows, Linux, Mac)
- High performance, async I/O
- Can run **standalone** or with **reverse proxy**

```
var builder = WebApplication.CreateBuilder();  
var app = builder.Build();  
// Kestrel starts here  
app.Run();
```



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WHAT IS IIS?

IIS (Internet Information Services) is Microsoft's web server for Windows. With .NET Core, it acts as a reverse proxy in front of Kestrel.

- **Windows-only** web server
- Battle-tested and enterprise-ready
- Rich management tools
- Acts as **reverse proxy** for Kestrel



```
Internet → IIS (Port 80/443)
          → Kestrel (Port 5000)
          → Your .NET Core App
```



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HOW THEY WORK TOGETHER

IIS receives requests and forwards them to Kestrel. **Kestrel** processes the request and sends response back through IIS.

Flow: Client → IIS (handles SSL, logging) → Kestrel (runs your app) → Response

```
// IIS Integration in Program.cs
builder.WebHost.UseIISIntegration();

// IIS forwards to Kestrel
// Request: IIS:443 → Kestrel:5000
// Response: Kestrel → IIS → Client
```



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REVERSE PROXY PATTERN

A **reverse proxy** sits in front of your app server, handling external requests and forwarding them internally.

Benefits:

- SSL termination
- Load balancing
- Security layer
- Static file serving



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WHEN TO USE KESTREL ALONE

Best Use Cases:

- Linux/Mac deployments where IIS isn't available
- Docker containers for simplified setup
- Microservices behind API gateway
- Development for fast iteration

```
• • • +  
  
# Linux production with Kestrel only  
dotnet MyApp.dll --urls "http://*:5000"  
  
# Usually behind Nginx reverse proxy  
nginx → Kestrel (your app)|
```



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WHEN TO USE IIS + KESTREL

Best Use Cases:

- Windows Server production environment
- Enterprise requirements for IIS features
- Existing IIS infrastructure
- Advanced logging and monitoring needs

```
←!— web.config for IIS —→  
<aspNetCore processPath="dotnet"  
  arguments=". \MyApp.dll"  
  stdoutLogEnabled="true">  
</aspNetCore>
```



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KEY DIFFERENCES

- Platform:
 - Kestrel is cross-platform
 - IIS is Windows-only
- Role:
 - Kestrel runs your app
 - IIS acts as reverse proxy
- Deployment:
 - Kestrel can run alone
 - IIS always uses Kestrel internally



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